

BillBull Retail OS

Financial Flow Architecture

Developer Reference Document

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Sales & Tax • Procurement & Inventory • Financial Reports

01 Financial Architecture Principles

BillBull Retail OS is built on a double-entry accounting engine where every financial event creates a balanced pair of debit and credit postings. Before any module is built, the team must internalize these foundational rules because every screen, every API endpoint, and every background job ultimately traces back to them.

The Double-Entry Foundation

At its core, BillBull enforces that for every transaction: $\text{Total Debits} = \text{Total Credits}$. There are no exceptions. A Sales Invoice simultaneously debits the customer receivable and credits the revenue and VAT payable accounts. A Purchase Invoice simultaneously credits the supplier payable and debits the inventory and input VAT accounts. The system must never allow a posting that leaves these totals unbalanced.

The Accounting Equation

$\text{Assets} = \text{Liabilities} + \text{Equity}$. Every transaction in BillBull must respect this equation. When a sale is made on credit, Assets (Receivables) go up and Equity (Retained Earnings via Revenue) goes up equally. When stock is purchased, Assets (Inventory) go up and Liabilities (Payables) go up equally. The balance sheet must always balance.

Multi-Dimensional Posting

BillBull operates across branches, outlets, cost centers, and item batches. Every journal entry carries dimensional tags — branch code, outlet ID, cost center code — that allow the same transaction to simultaneously update the consolidated company books and the branch-level P&L and balance sheet. This is not optional; dimensions must be enforced at the posting engine level, not left to individual developers.

The Posting Engine Contract

The Posting Engine is the single gateway through which all accounting entries pass. No module should write directly to the ledger tables. The Posting Engine validates balance, assigns voucher numbers, attaches dimensions, records timestamps and user IDs, and then writes to both the transaction ledger and the general ledger summary. This ensures auditability and consistency across all flows.

Period Control and Locking

Financial periods must be lockable. Once a month-end close is performed, no entry should be permitted in that period. The system must enforce period status checks at the Posting Engine level. All backdated entries must be restricted by user permission level. A period lock effectively protects the integrity of all financial reports for that period.

Key Architecture Rules

- Every module routes postings through the central Posting Engine — no exceptions

- Dimensions (Branch, Outlet, Cost Center) are mandatory on every journal line
- Period lock is enforced before writing any entry
- A unique, sequential voucher number is assigned per transaction type per branch
- All postings are soft-deletable with reversal entries — no hard deletes of accounting records
- Tax entries (VAT input/output) are tracked in dedicated sub-ledger accounts
- Foreign currency transactions maintain both local and foreign currency amounts in the ledger

02 Core Accounting Masters

Masters are the configuration backbone of the financial engine. Getting them right from the start prevents structural problems that are very costly to fix later. Developers must build the master data module before any transactional module.

Fiscal Year Master

Each fiscal year defines the start date, end date, and period structure (monthly, quarterly). The system must support overlapping fiscal year setup for year-end transition. A fiscal year can have a status of Open, Closing, or Closed. Postings are only allowed in Open periods. Developers must enforce a date-range check against the active fiscal year before allowing any posting.

Currency Master

BillBull supports multi-currency operations. The Currency Master defines the base/functional currency (typically AED or OMR), each foreign currency code, and exchange rate history. Exchange rates must be dated — the rate effective on the transaction date must be used for conversion. The system stores both the foreign currency amount and the base currency equivalent on every transaction line.

Tax Master (VAT)

The Tax Master stores each tax code, the tax rate (e.g. 5%), the applicable input tax account, and the output tax account. For UAE VAT, the standard rate is 5% but zero-rated and exempt categories must also be configured. The tax code is linked to the item master and the customer/vendor master to determine the default VAT treatment on each transaction line.

Payment Terms Master

Payment terms define how due dates are calculated on invoices (e.g. Net 30, 2/10 Net 30 for early payment discount). These are linked to customer and vendor masters and drive the aging reports. The early payment discount percentage is also stored here and referenced by the receipts and payments module.

Customer Master

Each customer record carries a linked accounts receivable sub-ledger account, default payment terms, default VAT category, credit limit, and currency. The customer master is the anchor point for all sales

transactions. The AR sub-ledger balance must always reconcile to the AR control account in the general ledger.

Vendor Master

Each vendor record mirrors the customer master but for the payables side. It carries the linked AP sub-ledger account, default payment terms, default VAT category (for input tax), bank details for payment processing, and currency. The AP sub-ledger must reconcile to the AP control account.

Employee Master

The Employee Master stores salary structure (basic pay, allowances, deductions), the linked salary payable account, advance ledger account, and bank details for WPS payments. This is referenced by the payroll module to generate salary journals and by HR for leave and loan tracking.

03 Chart of Accounts Structure

The Chart of Accounts (COA) is the numbering scheme for all accounts. BillBull uses a structured five-group hierarchy. Account numbers must be designed to allow future expansion without restructuring.

Account Group	Number Range	Nature	Balance Sheet / P&L
Assets	1000 – 1999	Debit Normal	Balance Sheet
Current Assets	1000 – 1299	Debit Normal	Balance Sheet
Fixed Assets	1300 – 1599	Debit Normal	Balance Sheet
Liabilities	2000 – 2999	Credit Normal	Balance Sheet
Current Liabilities	2000 – 2399	Credit Normal	Balance Sheet
Long Term Liabilities	2400 – 2999	Credit Normal	Balance Sheet
Equity	3000 – 3999	Credit Normal	Balance Sheet
Revenue	4000 – 4999	Credit Normal	P&L
Cost of Goods Sold	5000 – 5999	Debit Normal	P&L
Operating Expenses	6000 – 6999	Debit Normal	P&L
Other Income	7000 – 7499	Credit Normal	P&L
Other Expenses	7500 – 7999	Debit Normal	P&L

Key Account Codes — BillBull Standard

Account Name	Code	Type
Cash in Hand	1001	Asset – Current
Bank Account (Main)	1010	Asset – Current
Bank Account (Collection)	1011	Asset – Current

Accounts Receivable Control	1100	Asset – Current
Customer Advances Received	2050	Liability – Current
Inventory – Raw / Retail	1200	Asset – Current
Inventory – In Transit	1201	Asset – Current
VAT Input Tax	1310	Asset – Current
Prepaid Expenses	1320	Asset – Current
Fixed Assets – Equipment	1400	Asset – Fixed
Accumulated Depreciation	1450	Asset – Contra
Accounts Payable Control	2001	Liability – Current
Vendor Advances Paid	1105	Asset – Current
VAT Output Tax	2100	Liability – Current
VAT Payable (Net)	2101	Liability – Current
Salary Payable	2200	Liability – Current
Salary Advances – Employees	1106	Asset – Current
Owner's Equity / Share Capital	3001	Equity
Retained Earnings	3100	Equity
Sales Revenue	4001	Revenue
Sales Returns	4002	Revenue – Contra
Trade Discounts Given	4003	Revenue – Contra
Purchase / COGS	5001	COGS
Purchase Returns	5002	COGS – Contra
Rent Expense	6001	Operating Expense
Utility Expense	6002	Operating Expense
Salary Expense	6010	Operating Expense
Discount Allowed (Sales)	6050	Operating Expense
Discount Received (Purchase)	7001	Other Income
Bank Charges	7501	Other Expense

04 Posting Engine Workflow

The Posting Engine is the most critical component in BillBull's financial architecture. Every transaction — whether it originates from the POS screen, a purchase order, a salary run, or a manual journal — must pass through this engine before any accounting record is written. Think of it as the financial gatekeeper.

Step-by-Step Flow of the Posting Engine

Step 1 — Transaction Initiation

A user action or a system trigger (e.g. confirming a Sales Order, approving a Purchase Invoice, running payroll) generates a posting request. This request is a structured object containing: transaction type code, document reference number, transaction date, line items with account codes, amounts in both debit and credit columns, dimensional tags (branch, outlet, cost center), and the user ID and timestamp.

Step 2 — Pre-Validation

The engine validates: (a) the fiscal period is open for the transaction date, (b) all account codes exist and are active, (c) all mandatory dimensions are present, (d) foreign currency amounts have a valid exchange rate for the transaction date, and (e) credit limits have not been breached for customer transactions. If any check fails, the transaction is rejected with a descriptive error code and no entry is written.

Step 3 — Balance Check

The engine sums all debit amounts and all credit amounts across all lines of the transaction. If Total Debits $\neq$ Total Credits, the transaction is rejected. This check is absolute and cannot be bypassed. Even rounding differences (e.g. from currency conversion) must be absorbed into a designated rounding account before the entry is submitted.

Step 4 — Voucher Number Assignment

The engine assigns a unique, sequential voucher number using a per-branch, per-transaction-type sequence. For example: SI-DXB-2024-000123 for Sales Invoice number 123 at the Dubai branch in 2024. This number is locked and never reused, even if the transaction is later reversed. Sequence tables must be row-locked during assignment to prevent duplicates in concurrent environments.

Step 5 — Ledger Write

The engine writes two sets of records simultaneously within a database transaction. First, the transaction ledger (detailed entries with all line-level data, dimensions, and source document reference). Second, it updates the general ledger running balance table by account, period, and dimension combination. Both writes must succeed atomically — if either fails, the entire database transaction is rolled back.

Step 6 — Sub-Ledger Update

For transactions affecting customers or vendors, the engine also writes to the respective AR or AP sub-ledger. For inventory transactions, it writes to the inventory movement ledger. These sub-ledgers feed the aging reports and stock valuation reports without requiring full ledger recalculation.

Step 7 — Notification and Status Update

On success, the engine updates the source document status (e.g. Sales Order status changes from 'Confirmed' to 'Posted'), triggers any downstream workflow (e.g. dispatch queue), and logs the posting in the audit trail. The audit trail records: voucher number, source document, posting user, posting timestamp, and system IP/session.

Posting Engine: What It MUST NOT Do

- It must never partially post — all lines or no lines (atomicity is non-negotiable)
- It must never allow posting to a locked period, even for system-generated entries

- It must never generate duplicate voucher numbers — use database-level sequence locking
- It must never bypass dimension validation — incomplete dimensions corrupt reports
- It must never allow a manual override of the balance check

05 Sales Flow — Sales Order (SO)

The Sales Order flow is the primary revenue-generating process in BillBull. A sale can originate from the POS terminal (retail walk-in), a counter sale, or a business-to-business credit sale. The flow moves from order creation through invoicing to payment collection, with each step generating precise accounting entries.

Phase 1 — Sales Order Creation

The salesperson or cashier creates a Sales Order by selecting a customer (or the default walk-in customer for cash sales), adding line items with quantities and unit prices. At this stage, no accounting entry is posted. The SO is a commercial document. However, the system must check the customer's credit limit and alert if the limit will be breached. The system also pre-calculates VAT on each line based on the item's tax code.

Phase 2 — Sales Invoice Generation

When the SO is confirmed and goods are ready to dispatch (or in the case of a POS sale, at checkout), the system generates a Sales Invoice (SI). This is the trigger for the accounting Posting Engine. The invoice carries the customer name, invoice date, payment terms, line items with quantities, unit prices, line discounts, line VAT amounts, and the invoice totals.

Posting: Credit Sale Invoice

Account	Debit (Dr)	Credit (Cr)	Report Impact
Accounts Receivable (Customer)	Invoice Total		BS – Asset ↑
Sales Revenue		Net Sales Amount	P&L – Revenue ↑
VAT Output Tax		VAT Amount	BS – Liability ↑

Posting: Cash Sale Invoice (POS)

Account	Debit (Dr)	Credit (Cr)	Report Impact
Cash in Hand / Card Collection Account	Invoice Total		BS – Asset ↑
Sales Revenue		Net Sales Amount	P&L – Revenue ↑
VAT Output Tax		VAT Amount	BS – Liability ↑

Simultaneous Inventory Posting (COGS)

At the same moment the Sales Invoice is posted, the system also posts the Cost of Goods Sold entry. The item's weighted average cost (or batch cost) is fetched from the inventory ledger, and the following entry is made:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Cost of Goods Sold (COGS)	Cost Amount		P&L – Expense ↑
Inventory Account		Cost Amount	BS – Asset ↓

Report Impacts — Sales Invoice

Financial Report	Impact
Balance Sheet	Receivables (or Cash) increases by invoice total; Inventory decreases by cost; VAT Output Liability increases
Profit & Loss	Revenue increases by net sales amount; COGS increases by item cost; Gross Profit margin is created
Trial Balance	AR Dr, Revenue Cr, VAT Output Cr, COGS Dr, Inventory Cr — all balanced
Cash Flow (Indirect)	No direct cash impact for credit sales; Cash collected appears in Operating Activities when received
VAT Return	Output VAT accumulates in the VAT Output account for the tax return period
Customer Ledger	Customer's outstanding balance increases; feeds AR Aging report

06 Customer Advance Against SO

When a customer pays an advance before the Sales Order is fully delivered or invoiced, BillBull must handle this as a liability (unearned revenue) — not as income. The advance is held in the Customer Advances account until the invoice is raised and the advance is applied.

Step 1 — Receiving the Advance

The customer pays a deposit or partial payment against a future order. The cashier records this as a Customer Advance Receipt. At this point, no revenue is recognized — the money received is a liability because the goods/services have not yet been delivered.

Account	Debit (Dr)	Credit (Cr)	Report Impact
Cash / Bank / Card Account	Advance Amount		BS – Asset ↑
Customer Advances (Liability)		Advance Amount	BS – Liability ↑

Step 2 — Applying Advance to Invoice

When the Sales Invoice is raised and the advance is applied as full or partial payment, the Customer Advance liability is cleared against the Accounts Receivable. If the advance fully covers the invoice, no outstanding receivable remains. If it partially covers it, the remaining balance stays in AR.

Account	Debit (Dr)	Credit (Cr)	Report Impact
Customer Advances (Liability)	Advance Applied		BS – Liability ↓
Accounts Receivable		Advance Applied	BS – Asset ↓

► At the time the Sales Invoice is posted, the full AR entry is created first (DR Receivables, CR Revenue, CR VAT Output). The advance application entry then separately reduces the AR balance. Net AR = Invoice Total – Advance Applied.

Step 3 — Advance Refund (if order is cancelled)

If the order is cancelled before delivery and the advance must be refunded, the liability is cleared and cash is paid out:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Customer Advances (Liability)	Refund Amount		BS – Liability ↓
Cash / Bank Account		Refund Amount	BS – Asset ↓

Important: Advance vs Revenue

- Customer Advances are NEVER posted to Revenue at the time of receipt
- Revenue is only recognized when the invoice is raised (goods/services delivered)
- The advance sits in a current liability account until matched to an invoice
- The Customer Ledger must show both the advance and the invoice separately until matched
- Unmatched advances on the balance sheet date must be disclosed as current liabilities

07 Discounts and VAT

Discounts and VAT are two of the most frequently mishandled areas in retail accounting systems. BillBull must calculate VAT on the net amount after trade discounts but must handle settlement discounts differently. Understanding the distinction is critical.

Trade Discounts (Line Discounts)

A trade discount is a reduction in the selling price given at the time of sale — for example, 10% off the list price. VAT is calculated on the price AFTER the trade discount is applied. The discount itself is not a separate accounting entry; it simply reduces the net sale amount on the invoice line.

Example: Item list price = AED 100, Trade Discount = 10%, Net Price = AED 90, VAT 5% = AED 4.50, Invoice Total = AED 94.50

Account	Debit (Dr)	Credit (Cr)	Report Impact
Accounts Receivable / Cash	AED 94.50		BS – Asset ↑
Sales Revenue		AED 90.00	P&L – Revenue (net)
VAT Output Tax		AED 4.50	BS – Liability ↑

► For trade discounts, no separate Discount Allowed expense account is used. Revenue is simply recorded net of the discount.

Settlement / Cash Discounts (On Receipt)

A settlement discount is offered to a customer for early payment (e.g. 2% if paid within 10 days). This is recorded as an expense (Discount Allowed) at the time of payment, not at the time of invoicing. The original invoice is posted at full value. See Section 12 for full treatment under Receipts and Payments.

VAT Calculation Rules

Scenario	VAT Base	Treatment
Standard rated item with no discount	Full selling price	5% on full price
Standard rated item with trade discount	Price after discount	5% on net price
Zero-rated item (e.g. basic food)	N/A	0% — still needs VAT line with zero amount for reporting
Exempt item	N/A	No VAT charged, no input VAT recovery on related costs
Mixed invoice (standard + zero rated)	Per line	Calculate VAT per line; sum totals
Settlement discount (not yet taken)	Full invoice amount	VAT on full amount; adjust only if discount is actually taken

VAT Posting Detail

The VAT Output account (2100) accumulates all VAT collected from customers during the period. The VAT Input account (1310) accumulates all VAT paid to vendors on purchases. At the time of the VAT return filing, the net VAT payable (Output VAT minus Input VAT) is transferred to the VAT Payable account (2101) and subsequently paid to the tax authority.

Account	Debit (Dr)	Credit (Cr)	Report Impact
VAT Output Tax (2100)	Output VAT Total		<i>Clears output account</i>
VAT Input Tax (1310)		Input VAT Total	<i>Clears input account</i>
VAT Payable (2101)		Net VAT Due	<i>BS – Liability (net payable)</i>

08 Sales Return / Credit Note

When a customer returns goods, BillBull creates a Credit Note (CN) against the original Sales Invoice. The Credit Note reverses the revenue, reverses the VAT output, and increases the inventory back to stock. The CN must always reference the original invoice for audit trail purposes.

Step 1 — Goods Return to Stock

Before the financial entry, the warehouse confirms goods are received back and inspected. The inventory module creates a Goods Return Note which triggers the inventory entry and the COGS reversal.

Step 2 — Credit Note Posting

Account	Debit (Dr)	Credit (Cr)	Report Impact
Sales Returns (Contra Revenue) / Revenue	Net Return Amount		P&L – Revenue ↓
VAT Output Tax	VAT on Return		BS – Liability ↓
Accounts Receivable / Cash		Total Returned Amount	BS – Asset ↓

Inventory Reversal (COGS Reversal)

Account	Debit (Dr)	Credit (Cr)	Report Impact
Inventory Account	Original Cost of Returned Goods		BS – Asset ↑
Cost of Goods Sold		Original Cost of Returned Goods	P&L – COGS ↓

► The cost used for the COGS reversal must be the original cost at which the item was sold, not the current average cost. This prevents inventory valuation distortions.

Applying Credit Note to Outstanding Balance

Once the CN is raised, it can be applied against the customer's open invoices. The application entry simply reduces both the AR balance and the CN balance simultaneously, without any further P&L impact. If cash is refunded instead, a payment entry is made.

Account	Debit (Dr)	Credit (Cr)	Report Impact
Accounts Receivable (CN applied to invoice)		CN Amount	Reduces open AR
Accounts Receivable (original invoice)	CN Amount applied		Clears invoice balance

Report Impacts — Credit Note

Report	Impact
P&L	Net Revenue decreases; COGS decreases proportionally
Balance Sheet	AR decreases (or Cash decreases if cash refund); Inventory increases; VAT Output Liability decreases
VAT Return	Output VAT is reduced — must be reported in the relevant VAT return period
Customer Ledger	CN offsets open invoice balance; reduces customer outstanding
Inventory	Stock quantity and value both increase at original cost

09 Purchase Flow — LPO to Invoice

The purchase flow starts with a Local Purchase Order (LPO) and flows through goods receipt to vendor invoice posting. The key principle is that the accounting entry for the purchase liability is triggered by receipt of goods or the vendor invoice (whichever is the policy), not by the LPO.

Phase 1 — Local Purchase Order (LPO)

The LPO is a commitment document sent to the vendor. At the time of LPO creation, NO accounting entry is posted. The LPO is a commercial commitment only. However, the system must track open LPOs for the Commitment Report, which shows outstanding purchase commitments for cash flow forecasting.

Phase 2 — Goods Receipt Note (GRN)

When goods arrive at the warehouse, the storekeeper creates a Goods Receipt Note (GRN) against the LPO. At this point, the inventory enters the system. If BillBull follows the 3-way matching approach (LPO + GRN + Invoice), the liability is provisionally recorded when the GRN is created. This is called an Accrued Liability or Goods Received Not Invoiced (GRNI) entry:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Inventory Account	GRN Value (at PO cost)		BS – Asset ↑
Goods Received Not Invoiced (GRNI) — Accrual		GRN Value	BS – Liability (temp)

Phase 3 — Vendor Invoice (Purchase Invoice)

When the vendor invoice arrives and is matched to the GRN, the GRNI accrual is reversed and the actual vendor payable is recorded:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Goods Received Not Invoiced (GRNI)	GRN Value		BS – Liability clears
VAT Input Tax	VAT Amount on Invoice		BS – Asset ↑ (recoverable)
Accounts Payable (Vendor)		Invoice Total (incl. VAT)	BS – Liability ↑

► If price differences exist between the GRN cost and the invoice cost, the variance is posted to a Purchase Price Variance account and investigated.

Report Impacts — Purchase Invoice

Report	Impact
Balance Sheet	AP Liability increases; Inventory already on BS from GRN; Input VAT Asset increases

P&L	No immediate P&L impact — COGS is recorded when goods are sold, not when purchased
Cash Flow	Payable created — cash outflow occurs when vendor is paid (financing/operating activity)
VAT Return	Input VAT is claimable — reduces net VAT payable
Inventory	Stock value and quantity confirmed at invoice cost
Vendor Ledger	Vendor outstanding balance increases; feeds AP Aging report

10 Vendor Advance Against LPO

Sometimes vendors require a partial or full advance payment before shipping goods. This advance must be tracked as a prepaid asset (money owed to the business in the form of goods yet to be delivered) and applied against the vendor invoice when goods are received.

Step 1 — Paying the Vendor Advance

Account	Debit (Dr)	Credit (Cr)	Report Impact
Vendor Advances Paid (Asset)	Advance Amount		BS – Asset ↑
Bank / Cash Account		Advance Amount	BS – Asset ↓

Step 2 — Applying Advance to Purchase Invoice

When the vendor invoice is posted and the advance is applied, the advance asset is cleared against the accounts payable:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Accounts Payable (Vendor)	Advance Applied		BS – Liability ↓
Vendor Advances Paid (Asset)		Advance Applied	BS – Asset ↓

Step 3 — Advance Refund (if order cancelled)

Account	Debit (Dr)	Credit (Cr)	Report Impact
Bank / Cash Account	Refund Amount		BS – Asset ↑
Vendor Advances Paid (Asset)		Refund Amount	BS – Asset ↓

Advance Management Rules

- Vendor Advances must NEVER be posted directly to Purchases/Expenses
- Each advance must reference a specific LPO number for traceability
- Advances outstanding at period-end appear on the Balance Sheet as a current asset (amounts owed by vendors)

- The AP sub-ledger must show advance amounts separately from invoice amounts
- VAT on advances (where applicable) must follow local tax authority rules on when input VAT can be claimed

11 Purchase Return / Debit Note

When goods are returned to a vendor (due to defects, wrong delivery, or over-shipment), BillBull raises a Debit Note (DN) which reduces the vendor payable, reverses the inventory, and reverses the input VAT that was previously claimed.

Purchase Return Posting

Account	Debit (Dr)	Credit (Cr)	Report Impact
Accounts Payable (Vendor)	Return Amount (incl. VAT)		BS – Liability ↓
Inventory Account		Return Cost	BS – Asset ↓
VAT Input Tax		VAT on Return	BS – Asset ↓ (claim reversed)

Applying the Debit Note

The Debit Note reduces the vendor's outstanding balance and can be either (a) applied against a future invoice from the same vendor, or (b) received as a cash refund. If received as cash:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Bank / Cash Account	Refund Amount		BS – Asset ↑
Accounts Payable (Vendor)		Refund Amount	Clears DN balance

► Inventory adjustments for returns must use the original GRN cost, not the current average cost, to prevent distortion of the weighted average cost calculation.

12 Receipts and Payments

Receipts and Payments represent the cash and bank movement layer of the financial system. Every time money moves — whether collecting from a customer or paying a vendor — the system must create a precise accounting entry. Settlement discounts offered at the time of receipt or payment require special handling.

Customer Receipt — Full Payment

Account	Debit (Dr)	Credit (Cr)	Report Impact
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Bank / Cash Account	Amount Received		BS – Asset ↑
Accounts Receivable		Amount Received	BS – Asset ↓

Customer Receipt — With Settlement Discount

When a customer pays early and takes the offered early payment discount, the invoice is fully settled even though less cash was received. The discount amount is recognized as an expense (Discount Allowed):

Account	Debit (Dr)	Credit (Cr)	Report Impact
Bank / Cash Account	Cash Received (net)		BS – Asset ↑
Discount Allowed (Expense)	Discount Amount		P&L – Expense ↑
Accounts Receivable		Full Invoice Amount	BS – AR fully cleared

► *VAT Note: In some jurisdictions (including UAE), if a settlement discount is granted, the output VAT must be adjusted to reflect the reduced consideration. BillBull must support a VAT adjustment entry linked to the receipt if local tax rules require it.*

Vendor Payment — Full Payment

Account	Debit (Dr)	Credit (Cr)	Report Impact
Accounts Payable (Vendor)	Amount Paid		BS – Liability ↓
Bank / Cash Account		Amount Paid	BS – Asset ↓

Vendor Payment — With Settlement Discount

When BillBull takes advantage of a vendor's early payment discount, the discount is recognized as income (Discount Received):

Account	Debit (Dr)	Credit (Cr)	Report Impact
Accounts Payable (Vendor)	Full Invoice Amount		BS – Liability fully cleared
Bank / Cash Account		Cash Paid (net)	BS – Asset ↓
Discount Received (Income)		Discount Amount	P&L – Other Income ↑

Post-Dated Cheque Receipt

When a customer gives a post-dated cheque, it must not be deposited into the Bank account immediately. It goes into a Cheques Under Collection (CUC) account until the clearing date:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Cheques Under Collection (CUC)	Cheque Amount		BS – Asset (interim)
Accounts Receivable		Cheque Amount	BS – AR ↓

On the clearing date when the bank confirms credit:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Bank Account	Cheque Amount		BS – Bank ↑
Cheques Under Collection (CUC)		Cheque Amount	BS – CUC ↓

13 Salary Payments, Advances & Deductions

The payroll module handles monthly salary processing, advance payments during the month, and end-of-month deductions. All flows must maintain linkage between the salary journal and the individual employee sub-ledger.

Salary Advance During the Month

When an employee requests a salary advance, the system pays out cash and records a receivable from the employee (tracked in the Salary Advances account):

Account	Debit (Dr)	Credit (Cr)	Report Impact
Salary Advances — Employee (Asset)	Advance Amount		BS – Asset ↑
Cash / Bank Account		Advance Amount	BS – Asset ↓

Monthly Salary Journal (End of Month)

At month-end, the payroll run creates a salary expense journal for the total salary obligation. This includes basic salary, all allowances, and employer social insurance contributions. Gross salary is the total before deductions:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Salary Expense (Operating Expense)	Gross Salary Amount		P&L – Expense ↑
Salary Payable (Liability)		Net Payable to Employee	BS – Liability ↑
Salary Advances — Employee		Advance Deducted	BS – Asset ↓ (cleared)
Other Deductions Payable		Other Deductions	BS – Liability ↑

WPS Payment (Salary Disbursement)

When the bank transfer is made to employee accounts via the Wage Protection System:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Salary Payable (Liability)	Net Salary Transferred		BS – Liability ↓

Bank Account		Net Salary Transferred	BS – Asset ↓
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Report Impacts — Payroll

Report	Impact
P&L	Salary Expense increases by gross payroll amount each month
Balance Sheet	Salary Payable appears as current liability from month-end until payment date; Advances appear as current asset until deducted
Cash Flow	Salary payment is an operating cash outflow; recorded in the period of payment
Cost Center Report	Each employee's salary must be tagged to their department/cost center for departmental P&L

14 Journal, Contra & Expenses

Manual Journal Voucher

A Manual Journal is used for accounting entries that do not originate from a business transaction module — for example, depreciation, accruals, prepayments, and corrections. Every manual journal must have: a date, a narration, at least two lines, balanced debits and credits, a cost center tag, and an approver. The system must enforce maker-checker approval for manual journals above a defined threshold.

Depreciation Journal (Monthly)

Account	Debit (Dr)	Credit (Cr)	Report Impact
Depreciation Expense	Monthly Depreciation		P&L – Expense ↑
Accumulated Depreciation		Monthly Depreciation	BS – Contra Asset ↑

Prepaid Expense Amortization

Account	Debit (Dr)	Credit (Cr)	Report Impact
Expense Account (e.g. Rent Expense)	Period Portion		P&L – Expense recognized
Prepaid Expenses (Asset)		Period Portion	BS – Prepaid Asset ↓

Accrued Expense (Month-End)

Account	Debit (Dr)	Credit (Cr)	Report Impact
Expense Account	Accrued Amount		P&L – Expense ↑

Accrued Liabilities		Accrued Amount	BS – Liability ↑
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Contra Voucher

A Contra Voucher is used for cash-to-bank or bank-to-cash transfers within the same company. It is a balanced entry within the asset accounts and has no P&L impact. Common uses: depositing daily cash sales into the bank, or withdrawing petty cash from the bank.

Account	Debit (Dr)	Credit (Cr)	Report Impact
Bank Account	Transfer Amount		BS – Bank ↑
Cash in Hand		Transfer Amount	BS – Cash ↓

Direct Expense Voucher

For cash expenses paid directly from petty cash (e.g. office supplies, courier fees):

Account	Debit (Dr)	Credit (Cr)	Report Impact
Expense Account (e.g. Office Supplies)	Expense Amount		P&L – Expense ↑
Cash in Hand / Bank Account		Expense Amount	BS – Asset ↓

15 Bank, Card, Cheque and Reconciliation

BillBull handles multiple payment methods — cash, card (POS terminal), bank transfer, and cheques. Each method has its own clearing cycle and must be tracked through to bank statement reconciliation.

Card Payment (POS Terminal)

When a customer pays by debit or credit card, the money does not immediately reach the bank account — it goes through a payment gateway and clears within 1-3 business days. The interim account is the Card Settlement Receivable:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Card Settlement Receivable	Card Sale Amount		BS – Asset (pending)
Sales Revenue		Net Sales Amount	P&L – Revenue ↑
VAT Output Tax		VAT Amount	BS – Liability ↑

When the payment gateway settles to the bank (net of merchant fees):

Account	Debit (Dr)	Credit (Cr)	Report Impact
Bank Account	Net Settlement Amount		BS – Bank ↑
Bank Charges / Merchant Fee Expense	Merchant Fee		P&L – Expense ↑
Card Settlement Receivable		Gross Card Amount	BS – Clears receivable

Bank Reconciliation Process

The bank reconciliation matches the bank account balance in BillBull's books against the bank's official statement. Differences arise from timing (deposits in transit, outstanding cheques) or errors. The reconciliation must be performed monthly at minimum.

Reconciling Item	Book Treatment	Action Required
Deposit in transit	In books, not on statement	No action — will appear on next statement
Outstanding cheque issued	In books, not on statement	Monitor — void if stale
Bank charge not in books	On statement, not in books	Post a Bank Charges expense journal
Interest credited by bank	On statement, not in books	Post an Interest Income journal
Returned cheque (bounced)	On statement, not in books	Reverse receipt entry, notify customer
Error in books	Incorrect amount posted	Post a correcting journal with narration

► *BillBull should provide a Bank Reconciliation screen where the accountant marks each bank statement line as matched or unmatched, and the system shows the reconciled balance vs. book balance automatically.*

16 Inventory, Batch, Expiry and COGS

Inventory is one of the most complex areas in retail accounting. BillBull supports batch tracking and expiry date management, and must use a consistent cost flow method (Weighted Average Cost is recommended for retail) to calculate COGS accurately.

Inventory Valuation Method — Weighted Average Cost (WAC)

WAC recalculates the average cost per unit every time a new purchase receipt is posted. The formula is: $\text{New Average Cost} = (\text{Existing Stock Value} + \text{New Receipt Value}) / (\text{Existing Quantity} + \text{New Quantity})$. All subsequent sales are costed at this new average until the next receipt changes it.

Batch and Expiry Tracking

Each GRN line creates a batch record with: batch number, manufacture date, expiry date, received quantity, received cost, and current remaining quantity. When a sale is made for a batched item, the system must allocate from batches using FEFO (First Expiry First Out) to minimize wastage. The COGS entry uses the cost of the specific batch(es) consumed.

Inventory Movement Types and Postings

Movement Type	Inventory Account	Counter Account	P&L Impact
Purchase Receipt (GRN)	Dr Inventory	Cr GRNI / AP	None — cost stored as asset

Sale (Dispatch)	Cr Inventory	Dr COGS	COGS recognized on P&L
Purchase Return	Cr Inventory	Dr AP (Debit Note)	None — cost reversal
Sales Return	Dr Inventory	Cr COGS	COGS reduction
Stock Adjustment (Gain)	Dr Inventory	Cr Inventory Gain	Other Income ↑
Stock Adjustment (Loss)	Cr Inventory	Dr Inventory Loss	Other Expense ↑
Stock Transfer (Branch)	Dr Inventory – Receiving	Cr Inventory – Sending	None — internal
Expired / Written Off	Cr Inventory	Dr Inventory Write-off	Expense ↑

Expiry Management Postings

When stock is written off due to expiry:

Account	Debit (Dr)	Credit (Cr)	Report Impact
Inventory Write-off / Shrinkage Expense	Cost of Expired Goods		P&L – Expense ↑
Inventory Account		Cost of Expired Goods	BS – Inventory ↓

COGS and Gross Profit Flow

Gross Profit = Net Sales Revenue – Cost of Goods Sold. In BillBull, every sale posting simultaneously records both the revenue and the COGS. This means the Gross Profit figure on the P&L is always real-time and accurate. Developers must ensure that the inventory cost fetch (WAC or batch cost) and the revenue posting happen atomically in the same database transaction. If the inventory cost cannot be determined, the sale must not be permitted to post.

17 Financial Reports Mapping

All financial reports in BillBull are derived from the general ledger and sub-ledgers. Understanding which accounts feed which report is critical for developers building report modules. The following mapping defines the source accounts for each major report.

Balance Sheet Mapping

Balance Sheet Section	Accounts Included	Normal Balance
Current Assets	Cash (1001), Banks (1010-1019), Card Settlement Receivable, Cheques Under Collection, AR (1100), Inventory (1200), Prepaid (1320), Vendor Advances (1105), Employee Advances (1106), VAT Input (1310)	Debit

Fixed Assets	Equipment (1400), less Accumulated Depreciation (1450)	Debit (net)
Current Liabilities	AP (2001), Customer Advances (2050), Salary Payable (2200), VAT Output (2100), VAT Payable (2101), Accrued Liabilities	Credit
Equity	Share Capital (3001), Retained Earnings (3100), Current Year Net Profit	Credit

Profit & Loss Mapping

P&L Section	Accounts	Sign Convention
Gross Revenue	Sales Revenue (4001), less Sales Returns (4002)	Credit is positive
Cost of Goods Sold	COGS (5001), less Purchase Returns (5002)	Debit is cost
Gross Profit	Revenue minus COGS	Calculated
Operating Expenses	Salaries (6010), Rent (6001), Utilities (6002), Depreciation (6030), Discounts Allowed (6050), All 6xxx accounts	Debit is expense
EBITDA	Gross Profit minus Op. Expenses (before depreciation/interest)	Calculated
Other Income	Discount Received (7001), Interest Income (7002)	Credit is income
Other Expenses	Bank Charges (7501), Interest Expense (7502), Inventory Write-off	Debit is expense
Net Profit / (Loss)	All P&L accounts net	Calculated

Cash Flow Statement Mapping

BillBull generates the Cash Flow using the Indirect Method starting from Net Profit. Developers must maintain a classification flag on every account indicating whether it belongs to Operating, Investing, or Financing activities.

Cash Flow Section	Key Components
Operating Activities	Net Profit, add back Depreciation, changes in Working Capital (AR, Inventory, AP, Advances, Accruals), VAT net movements
Investing Activities	Purchase of Fixed Assets, disposal proceeds
Financing Activities	Owner capital injections, loan drawdowns, loan repayments, dividends paid
Net Change in Cash	Sum of all three sections — must reconcile to movement in Bank + Cash accounts

Trial Balance

The Trial Balance pulls every account's debit total and credit total for the selected period. The grand total of all debits must equal the grand total of all credits — this is the first-level validation of accounting integrity. BillBull must support both the summarized TB (one line per account) and the detailed TB (one line per journal entry). The TB is the source from which the Balance Sheet and P&L are compiled.

18 Branch, Outlet and Cost Center Dimensions

Dimensions allow BillBull to generate financial reports at multiple levels of granularity without maintaining separate chart of accounts for each location. Every transaction carries dimensional tags that enable multi-level reporting.

Dimension Hierarchy

Level	Entity	Purpose
Level 1 — Company	BillBull Retail LLC (or parent entity)	Consolidated financial statements
Level 2 — Branch	Dubai, Abu Dhabi, Sharjah, etc.	Branch P&L and Balance Sheet
Level 3 — Outlet	Mall outlet, standalone store within branch	Outlet-level Revenue and Expense
Level 4 — Cost Center	Sales, Operations, Admin, Warehouse	Departmental expense allocation
Level 5 — Item / Project	For specific project costing (if applicable)	Project-level profitability

How Dimensions Work in Postings

When the Dubai outlet raises a Sales Invoice, the posting engine tags all journal lines with Branch=DXB, Outlet=DXB-MALL-01, CostCenter=SALES. The general ledger stores this alongside the account code. This means the system can generate a P&L showing: All Companies → Dubai Branch → Mall Outlet → Sales Department, each level being a drill-down of the previous.

Inter-Branch Transactions

When inventory is transferred from one branch to another, or when head office costs are allocated to branches, the system uses Inter-Company/Inter-Branch accounts to keep each entity's books balanced. The sending branch credits its Inventory account and debits an Inter-Branch Receivable. The receiving branch debits its Inventory and credits an Inter-Branch Payable. These must offset each other at consolidation.

Dimension Enforcement Rules

- All posting lines must carry at minimum a Branch code — this is mandatory
- Outlet is mandatory for revenue and COGS lines
- Cost Center is mandatory for expense lines
- Missing dimensions are a data quality failure — reject the transaction, do not default silently
- Dimension combinations must be validated against a dimension matrix (not all combinations are valid)
- Reports must support filtering and grouping by any dimension or combination of dimensions

19 Developer Data Flow

This section describes the data flow architecture that developers must implement to ensure all transactions are correctly captured, validated, and posted to the general ledger. Understanding the data flow is essential before writing any transactional code.

Transaction Lifecycle States

State	Description	Can Post?
Draft	Transaction being entered by user — data incomplete or unconfirmed	No
Pending Approval	Submitted for manager/accountant approval — data locked	No
Approved	Approved and ready for posting — triggers Posting Engine	Yes
Posted	Accounting entries written to ledger — document locked	No (only reversal)
Partially Applied	Invoice partially paid or partially applied	No
Fully Applied / Settled	All balances cleared	No
Reversed	Original entries reversed by a reversal voucher	No
Cancelled	Document voided before posting — no accounting impact	No

Data Flow: Sales Invoice Example

User creates SO → System saves SO in Draft state → User confirms SO → System validates credit limit, stock availability, tax codes → SO moves to Approved state → System calls Posting Engine → Posting Engine validates period, balances, dimensions → Assigns voucher number → Writes to: (1) Transaction Ledger, (2) GL Summary, (3) AR Sub-ledger, (4) Inventory Movement Ledger, (5) VAT Sub-ledger → Updates SO status to Posted → Triggers dispatch queue → Returns success response to UI.

Table Architecture (Logical)

Table Name	Purpose	Key Columns
gl_transaction_header	One row per voucher	voucher_no, transaction_type, date, period_id, branch_id, ref_doc_no, status
gl_transaction_lines	One row per debit/credit line	header_id, account_id, debit_amt, credit_amt, branch_id, outlet_id, cost_center_id, currency, fx_rate, base_amt
gl_account_balances	Running balance per account+period+dimension	account_id, period_id, branch_id, outlet_id, debit_total, credit_total, closing_balance

ar_ledger	Customer open items	customer_id, voucher_no, doc_type, original_amt, applied_amt, outstanding_amt, due_date
ap_ledger	Vendor open items	vendor_id, voucher_no, doc_type, original_amt, applied_amt, outstanding_amt, due_date
inventory_movements	Stock movements with cost	item_id, batch_id, movement_type, quantity, unit_cost, total_cost, branch_id, posting_date
inventory_balances	Running stock balance	item_id, batch_id, branch_id, on_hand_qty, avg_cost, total_value

20 Indexing and Performance

Financial databases are read-heavy for reporting and write-heavy during business hours for transactions. Getting indexing right from day one prevents performance problems that become very hard to fix in production. These are the critical indexes and best practices for BillBull's accounting tables.

Critical Indexes

Table	Index	Reason
gl_transaction_header	Composite: (branch_id, transaction_date)	Date-range queries by branch for all reports
gl_transaction_header	Unique: voucher_no	Prevents duplicates, fast lookup by voucher
gl_transaction_header	Index: (ref_doc_no, transaction_type)	Trace from source document to journal
gl_transaction_lines	Composite: (account_id, period_id)	Account-level ledger queries
gl_transaction_lines	Composite: (header_id, line_no)	Join from header to lines
gl_transaction_lines	Composite: (branch_id, cost_center_id)	Dimension-filtered reports
gl_account_balances	Composite: (account_id, period_id, branch_id)	Fast running balance fetch
ar_ledger	Composite: (customer_id, due_date)	AR Aging report — most-queried
ar_ledger	Index: (outstanding_amt) WHERE outstanding_amt > 0	Partial index — open items only
ap_ledger	Composite: (vendor_id, due_date)	AP Aging report
inventory_movements	Composite: (item_id, branch_id, posting_date)	Stock history queries
inventory_balances	Composite: (item_id, batch_id, branch_id)	Real-time stock lookup
inventory_balances	Index: (expiry_date) WHERE expiry_date IS NOT NULL	Expiry-approaching alerts

Performance Best Practices

1. Pre-aggregate Balance Tables

Do not compute account balances by summing all transaction lines on every report request. The `gl_account_balances` table must be maintained as a real-time summary updated atomically with each posting. Reports read from this table, not from transaction lines. Transaction lines are only queried for drill-down detail.

2. Partition Large Tables by Period

The `gl_transaction_lines` table will grow very large in multi-branch retail environments. Partition it by fiscal year or by financial period. This ensures that a year-end report query scans only the current year's partition rather than all historical data.

3. Separate OLTP and OLAP Loads

Heavy financial reports (like detailed Trial Balance, full AR Aging, or Inventory Valuation Reports) should be run against a read replica of the database, not the primary transaction database. Schedule nightly or on-demand replication for the reporting replica.

4. Voucher Number Sequences

Use database-level sequences (not application-level counters with `SELECT MAX+1`). Application-level counters under concurrent load create duplicate voucher numbers. The sequence must be per-branch and per-transaction-type to keep numbers meaningful and compact.

5. Soft Delete Only

Never hard-delete accounting records. Use a status flag (`is_reversed`, `is_cancelled`) and reversal vouchers. This preserves the audit trail, prevents gap in voucher sequences, and allows forensic reconstruction of any balance at any point in time.

6. Period-End Lock at Database Level

Period locking must be enforced not just at the application layer but also optionally at the database layer using trigger-based controls. This prevents any batch script, ETL job, or direct SQL update from writing to a locked period.

7. Index Maintenance Schedule

Financial databases have high insert rates during business hours and high read rates during month-end reporting. Schedule index rebuilding and statistics updates during off-peak hours (early morning). Fragmented indexes on the AR Aging query can make a report that should take 2 seconds take 2 minutes.

21 QA Checklist

Before any financial module is released to production, the following test cases must pass. This checklist must be executed by a QA engineer who understands accounting principles, not just software testing.

A. Posting Engine

- Every posted transaction has Total Debits = Total Credits — verified by automated tests on every posting.
- Posting to a locked period is rejected with the correct error code.
- A transaction with a missing Branch dimension is rejected.
- Concurrent posting of two transactions does not produce duplicate voucher numbers.
- A database failure mid-write results in full rollback — no partial entries in the ledger.

B. Sales Flow

- Sales Invoice posting creates AR debit, Revenue credit, VAT Output credit — all correct amounts.
- COGS entry is created simultaneously with the invoice — Inventory credit equals COGS debit.
- A credit sale that exceeds the customer credit limit is blocked before posting.
- VAT is calculated on the net amount after trade discounts, not the list price.
- Cash sale at POS correctly debits Cash/Card and not AR.

C. Customer Advance

- Advance receipt does not post to Revenue — only to Customer Advances (Liability) and Cash/Bank.
- When advance is applied to an invoice, both AR and Customer Advances are reduced correctly.
- Advance refund correctly reduces both Cash and the Liability account.

D. Purchase Flow

- GRN creates inventory entry — no immediate P&L impact.
- Purchase Invoice clears GRNI and creates AP — input VAT is recorded as an asset.
- Price variance between GRN and Invoice is posted to the PPV account, not silently absorbed into inventory.

E. Returns

- Sales Credit Note reduces AR, reduces Revenue, reduces VAT Output, and restores Inventory at original cost.
- Purchase Debit Note reduces AP, reduces Inventory, and reduces VAT Input.
- Return entries reference the original invoice number in the audit trail.

F. Receipts / Payments with Discounts

- Receipt with settlement discount correctly debits Cash (net), debits Discount Allowed (expense), and credits AR (full invoice amount).
- Payment with settlement discount correctly credits Cash (net), credits Discount Received (income), and debits AP (full amount).

G. Payroll

- Salary journal Total Debit (Gross Salary Expense) = Total Credit (Net Payable + Advance Deducted + Other Deductions).
- WPS payment entry reduces Salary Payable to zero and reduces Bank by the net amount paid.
- Employee advance is visible on the Balance Sheet as a current asset until deducted.

H. Bank Reconciliation

- Bank account book balance matches sum of all bank account postings.
- Unreconciled items are clearly listed with aging.
- Bounced cheque reversal correctly re-opens the AR balance.

I. Inventory and COGS

- Weighted Average Cost recalculates correctly after every GRN.
- FEFO batch allocation correctly selects the earliest-expiring batch for sale.
- Inventory write-off for expired goods debits the write-off expense and credits inventory at correct cost.
- Stock transfer between branches keeps total company inventory value unchanged.

J. Financial Reports

- Balance Sheet total Assets = total Liabilities + Equity for any date.
- Trial Balance total Debits = total Credits for any period.
- P&L Net Profit equals the movement in Retained Earnings on the Balance Sheet.
- Cash Flow closing balance reconciles to Cash + Bank account balances on the Balance Sheet.
- AR Aging total equals AR Control Account balance on Balance Sheet.
- AP Aging total equals AP Control Account balance on Balance Sheet.
- Inventory Report total value equals Inventory account balance on Balance Sheet.

Final Note to the Development Team

Every item on this QA checklist represents a real-world accounting integrity failure mode. These are not hypothetical — they are based on actual bugs found in retail accounting systems. The goal of BillBull is to give every retailer — regardless of accounting knowledge — a system that produces financial statements they can trust and submit to auditors, banks, and tax authorities with confidence. The accounting engine is not a feature — it is the foundation.

Build it right. Test it thoroughly. Never compromise on the double-entry balance check.